

EPIC-AMP protein–peptide docking benchmark — results draft

Working draft for the thesis/paper Results section. Numbers are current as of the last pipeline run (2026-09-06); regenerate from results/REPORT.md, results/rerank.md, results/summary.csv before submission.

1. Aim and scope

We benchmark current all-atom structure-prediction methods on their ability to reproduce experimentally determined **protein–antimicrobial-peptide (AMP) complexes**, scored against the crystal structures with **DockQ**. Two questions:

- **Objective 1 — prediction accuracy.** How close is each method's top-ranked model to the native complex?
 - **Objective 2 — pose re-ranking.** Given a method's own set of models, can an external scoring function (GraphPep; InterPepRank) select a better pose than the method's own rank 1?
- DockQ is used only for evaluation, never for model selection.

2. Benchmark construction

2.1 Starting point

The seed set was a manually curated spreadsheet of protein–AMP crystal structures (AMP_data_edited.xlsx, ~18 entries) plus five AlphaFold-Multimer predictions from prior lab work. Every candidate had to be (i) a genuine antimicrobial peptide bound to (ii) a single protein partner, in (iii) a crystal structure suitable as a DockQ reference.

2.2 RCSB discovery sweep

To enlarge the set we ran an automated search of the RCSB PDB (pipeline/discover_targets.py): **26 AMP-family query terms** (defensin, cathelicidin, magainin, protegrin, LL-37, drosocin, apidaecin, pyrrhocoricin, thanatin, melittin, ...), resolution ≤ 2.8 Å, receptor 40–2000 aa, peptide 4–50 aa, exactly one receptor and one peptide chain. **506 unique candidates** were screened; **77** passed the geometric filter and were fully scored (results/discovery_candidates.csv): 39 ok, 13 caution, 25 drop by the automated resolved-fraction / non-standard-residue check.

Each surviving candidate was then **manually verified against the deposited entry's own citation title** to confirm real antimicrobial activity (not a crystal-form duplicate, not a protease-substrate or cytokine-antagonist false positive from the generic "antimicrobial peptide" term). Eight were added; four that had been added on a first pass were later removed (see 2.3 C–D).

2.3 Exclusion criteria

configs/targets.yaml is the single source of truth and retains every rejected entry (include: false) with its reason. **30 entries total → 13 included, 17 excluded.**

Class	Rule	n	Entries
A. Peptide disordered	< 50 % of the peptide modelled in the crystal → no reliable native interface	6	6Z2Q (2/19), 8GQA (9/20), 3QNJ (8/19), 4JWE (9/21), 8ITG (7/42), 6Y0X (5/12)
B. D-amino-acid / non-standard peptide	peptide chain contains D-residues or non-proteinogenic residues with no one-letter code	6	6Q6W, 6Q77, 6Q85, 6Q86, 6Y0W (all D-peptide ligands of LecB), 8ONU (DAB + hydroxyproline)
C. Unencodable residue in SEQRES	discovery candidate whose SEQRES carries a literal X (modified residue)	2	4JWI (PRPILLPWRX), 7NEF (KKLLKLLKLLX)

Class	Rule	n	Entries
D. Not a genuine AMP	discovery candidate whose citation shows a non-antimicrobial context	2	2HD4 (proteinase-K inhibitor peptide), 4N6P (lactoferrin fragment in a drug-binding study; also the representative of a 15-way duplicate cluster)
E. Resolved fragment not in contact	≥ 50 % modelled, but the ordered residues lie outside the receptor interface	1	4JWD (7/14 modelled; those 7 sit ~6.1 Å from the nearest receptor atom → DockQ finds no native interface)

Rationale for class B. Every method here (AlphaFold-Multimer, AlphaFold 3, Boltz-2, Chai-1, Protenix) takes a plain letter sequence and builds the L-enantiomer. Scoring an L-model against a D-native measures a mirror-image mismatch, not the predictor's skill — every method scores ≈ 0 identically, adding no signal — and an all-D chain crashes DockQ outright (Bio.PDB drops the all-non-standard chain, raising `KeyError`).

2.4 Final target set (n = 13)

8 ok (≥ 80 % of the peptide resolved), 5 caution (50–80 % resolved; DockQ computed over the modelled fragment, reported with an asterisk).

PDB	Receptor	Peptide	Len	Resolved	Quality	Source
6HY2	LpxA acyltransferase (*E. coli*)	peptide inhibitor	12	12/12	ok	seed
4E2S	DnaK chaperone (*E. coli*)	Metchnikowin	8	8/8	ok	seed
8GAL	LptA (*E. coli*)	Thanatin	20	18/20	ok	seed
6Z2P	O-glycan protease (*Akkermansia*)	Glycodrosocin	19	10/19	caution	seed
4JWC	DnaK (*E. coli*)	Cathelicidin-3 / Bac7-type	16	9/16	caution	seed
4EZR	DnaK (*E. coli*)	Drosocin	8	8/8	ok	discovery
4E81	DnaK (*E. coli*)	Apidaecin fragment	10	9/10	ok	discovery
4EZU	DnaK (*E. coli*)	PR-bombesin (*Bombina maxima*)	17	14/17	ok	discovery
4EZQ	DnaK (*E. coli*)	Pyrrhocoricin	9	7/9	caution	discovery
4EZO	DnaK (*E. coli*)	PR-39	15	8/15	caution	discovery
8AHT	Calmodulin	Melittin	26	25/26	ok	discovery
3QRX	Centrin	Melittin	26	20/26	caution	discovery
7SAY	engineered GCN4 scaffold	LL-37	37	35/37	ok	discovery

Composition caveats. DnaK is the receptor in **7 of 13** targets (proline-rich AMPs threaded through the substrate channel); melittin (GIGAVLKVLTTGLPALISWIKRKRQQ) is the peptide in **two** targets with different host proteins (8AHT, 3QRX). 7SAY's "receptor" is an artificial crystallisation scaffold, not a natural partner — included only to have an LL-37 complex at all.

2.5 Native reference preparation

pipeline/prep_native.py downloads the RCSB entry, keeps biological assembly 1, retains one receptor + one peptide chain, strips ligands / waters / alternate locations / hydrogens, and writes data/natives/<ID>_native.pdb. data/natives/_audit.csv records the resolved fraction that drives the ok/caution/drop call.

3. Methods and how they were run

No local GPU was available; all predictors were run through their public web portals (manual submission, no API) except Boltz-2, which requires a GPU batch job.

Method	Model	Where run	Targets done	Notes
AlphaFold-Multimer	AF2 for complexes	COSMIC ² (SDSC HPC), web	6	5 models/target, ranked by iptm+ptm
AlphaFold 3	AF3	AlphaFold Server, web	10	5 models, ranking_score
Chai-1	open AF3-class	lab.chaidiscovery.com, web (free)	13	5 models, aggregate_score
Protenix	ByteDance AF3 reproduction	protenix-server.com, web (free)	13	5 samples, ranking_score
Boltz-2	open biomolecular model	university HPC (GPU)	1	blocked on cluster scheduler/env; only 6HY2

pipeline/collect_predictions.py normalises every download to a 2-chain PDB with native chain IDs and extracts each tool's own confidence to results/confidences.csv. Seed = 1 where user-controllable; server defaults elsewhere.
43 predictions total.

4. Evaluation protocol

pipeline/run_dockq.py runs **DockQ v2** as DockQ <model> <native> --mapping <rec><pep>:<rec><pep> --allowed_mismatches 3 --short. CAPRI bands (standard thresholds): **incorrect** < 0.23 ≤ **acceptable** < 0.49 ≤ **medium** < 0.80 ≤ **high**. Results roll up through pipeline/report.py into results/REPORT.md and results/summary.csv.

5. Results — Objective 1 (prediction accuracy)

5.1 Per method

Method	n	mean DockQ	≥ acceptable	≥ medium	≥ high	mean confidence	Spearman(confidence, DockQ)
Boltz-2	1	0.928	1/1	1/1	1/1	0.96	–
Protenix	13	0.650	11/13	10/13	4/13	0.79	0.68
AlphaFold-Multimer	6	0.520	5/6	3/6	1/6	0.73	0.80
AlphaFold 3	10	0.495	8/10	5/10	2/10	0.74	0.71
Chai-1	13	0.448	7/13	6/13	2/13	0.56	0.73

Means are over **different target subsets** (see §7). On the **complete 13-target set**, only Chai-1 and Protenix have run every target: **Protenix leads clearly** (mean 0.650, 10/13 ≥ medium) over Chai-1 (0.448, 6/13 ≥ medium). Boltz-2's single point (6HY2, 0.93) is its best-case and not comparable.

5.2 Per target (best pose, DockQ)

* = caution (partial peptide).

Target	Len	af3	af-mult	boltz	chai	protenix	best
4E81	10	0.90	0.93	–	0.95	0.90	0.95 (high)
4EZR	8	0.93	–	–	0.92	0.91	0.93 (high)
4EZX	8	0.29	0.29	–	0.38	0.93	0.93 (high)
6HY2	12	0.29	0.39	0.93	0.21	0.69	0.93 (high)
8GAL	20	–	0.75	–	0.79	0.83	0.83 (high)
4EZQ*	9	–	0.59	–	0.21	0.79	0.79 (medium)
6Z2P*	19	0.14	0.17	–	0.07	0.77	0.77 (medium)
4EZO*	15	0.71	–	–	0.69	0.75	0.75 (medium)
4JWC*	16	0.67	–	–	0.69	0.65	0.69 (medium)
8AHT	26	0.40	–	–	0.55	0.66	0.66 (medium)
3QRX*	26	0.53	–	–	0.23	0.39	0.53 (medium)
4EZU	17	0.08	–	–	0.09	0.18	0.18 (incorrect)
7SAY	37	–	–	–	0.07	0.01	0.07 (incorrect)

- **4 targets solved to high accuracy** by at least one method (4E81, 4EZR, 4EZX, 6HY2). All are short (8–12 aa).

- **2 targets fail for every method:** 7SAY (LL-37, best 0.07; but the receptor is a synthetic scaffold — arguably a mis-specified target) and 4EZU (PR-bombesin, best 0.18, fnat ≈ 0 — likely wrong site/register).

- **Large between-method spread on nominally easy targets.** 4EZX (8-aa peptide):

AF3 / AF-Multimer / Chai-1 all ≈ 0.29 – 0.38 (acceptable) but Protenix 0.925. 6HY2: Boltz-2 0.93 and Protenix 0.69 vs AF3 / Chai-1 ≈ 0.2 – 0.29 . The choice of method matters more than the apparent difficulty of the target.

5.3 Confidence vs accuracy

Pooled **Spearman(confidence, DockQ)** = **0.69** over 41 predictions; per-method 0.68–0.80. Each tool's self-reported confidence is a **usable filter** for which predictions to trust, though not a precise accuracy estimate.

5.4 Determinants of difficulty

- **Peptide length:** Spearman(length, DockQ) = **−0.43** — longer peptides score worse. The four high-accuracy targets are all ≤ 12 aa; the two total failures are the two longest (17, 37 aa).
- **Partial resolution (caution targets):** no systematic penalty — several of Protenix's best scores are on caution targets (6Z2P 0.77, 4EZ0 0.75) — but these DockQ values are computed over a peptide fragment and are weaker evidence.
- **Receptor family:** the DnaK proline-rich-AMP complexes (7/13) are mostly solved to medium/high by Protenix; the two hardest are non-DnaK (7SAY scaffold, and 4EZU which is DnaK but with an unusual peptide).

6. Results — Objective 2 (pose re-ranking)

6.1 Setup

Each method emits 5 ranked models per target. We treat these as a decoy set: `pipeline/rerank_prep.py` explodes them to per-pose PDBs (rank 00 = the method's own top pose), `run_dockq.py` scores **every** pose, and `pipeline/rerank_eval.py` compares three selection strategies by the DockQ of the selected pose — **docker top-1** (baseline), **re-ranker top-1**, and **oracle** (best of the 5). **41 ensembles** (af3 $\times 10$, chai $\times 13$, protenix $\times 13$, af-multimer $\times 4$, boltz $\times 1$), 5 poses each. For methods where the whole complex moves between models, `pipeline/graphpep_prep.py` superposes each pose's receptor onto rank 00's frame.

6.2 Re-ranking headroom (oracle – docker top-1)

Method	n	docker top-1 (mean DockQ)	oracle (mean DockQ)	headroom
AlphaFold 3	10	0.495	0.672	+0.177
Chai-1	13	0.448	0.562	+0.114
AlphaFold-Multimer	4	0.551	0.633	+0.082
Protenix	13	0.650	0.665	+0.016
Boltz-2	1	0.928	0.928	+0.000
All	41	0.545	0.637	+0.092

A perfect re-ranker would raise the $\geq$ **medium** count from 24/41 $\rightarrow$ 29/41 and $\geq$ **high** from 10/41 $\rightarrow$ 13/41. The headroom is **highly method-dependent**:

- **AF3 and Chai-1 leave substantial accuracy on the table** — their rank 1 is often not their best pose. Extreme cases: af3/6Z2P 0.14 $\rightarrow$ 0.84 (pose #2), chai/4EZQ 0.21 $\rightarrow$ 0.82 (#4), af3/6HY2 0.29 $\rightarrow$ 0.85 (#4), chai/6Z2P 0.07 $\rightarrow$ 0.45.
- **Protenix is already near-oracle** (+0.016) — its own confidence ranking is effectively optimal on this set, so external re-ranking has almost nothing to add.

This sets the ceiling any re-ranker must approach to be useful.

6.3 GraphPep

GraphPep (Huang lab, HUST; *Nat. Mach. Intell.* 2026) — interaction-derived graph network + ESM-2 features; binding score $-\log(1 + \text{predicted_fnat})$, more negative = better pose. Package: Zenodo 10.5281/zenodo.17099863 (trained weights bundled); run locally on CPU (WSL); see `methods/graphpep/`. Reproduced the shipped example bit-for-bit. All 41 ensembles scored.

Method	n	docker top-1	oracle	GraphPep top-1	Δ vs docker	win / tie / loss	mean $\rho(\text{score}, \text{DockQ})$
AlphaFold 3	10	0.495	0.672	0.620	+0.125	2 / 8 / 0	+0.02
Chai-1	13	0.448	0.562	0.461	+0.013	2 / 11 / 0	+0.04
AlphaFold-Multimer	4	0.551	0.633	0.465	-0.086	0 / 3 / 1	0.00
Protenix	13	0.650	0.665	0.612	-0.038	1 / 11 / 1	-0.42
Boltz-2	1	0.928	0.928	0.908	-0.019	0 / 0 / 1	-0.10
All	41	0.545	0.637	0.559	+0.014	5 / 33 / 3	-0.12

(win/tie/loss = GraphPep top-1 vs docker top-1, tie = within 0.02 DockQ. ρ = Spearman(GraphPep score, DockQ) within one 5-pose ensemble, per-method mean.)

GraphPep does not improve on the dockers' own ranking on this set. Overall it is within noise of the baseline (+0.014 DockQ, 33/41 ties) and recovers only ~15 % of the +0.092 headroom. The picture by method:

- **AlphaFold 3 — clear benefit (+0.125, 71 % of AF3's headroom recovered).**

Two decisive corrections drive it: af3/6Z2P 0.14 → 0.84 and af3/6HY2 0.29 → 0.85, in both cases selecting the true oracle pose. No losses. But it still misses af3/4EVS (leaves +0.39 on the table).

- **Chai-1 — neutral (+0.013).** Catches chai/6HY2 (0.21 → 0.37) and chai/4EVS

(0.375 → 0.410), ties everything else, and misses the two largest Chai headrooms (chai/4EZQ 0.21 → 0.82, chai/6Z2P 0.07 → 0.45).

- **Protenix and AlphaFold-Multimer — net harmful (-0.038, -0.086).** GraphPep

confidently swapped in a much worse pose on protenix/4EZQ (0.79 → 0.23) and afmultimer/4EZQ (0.59 → 0.23). Because Protenix's own ranking is already near-oracle, any re-ranking there is downside risk only.

- **Within-ensemble discrimination is essentially absent.** Pooled

Spearman(GraphPep score, DockQ) = **-0.12**; -0.42 for Protenix. GraphPep's score ordering does not track DockQ across a method's five models — it lands the right pose occasionally but not systematically.

- **CAPRI bands:** ≥ high 10 → 12/41 (the two AF3 saves); ≥ acceptable and ≥ medium

unchanged (31/41, 24/41).

Interpretation / caveats. GraphPep was trained on classical docking decoys (FlexPepDock, ADCP, LEADS-PEP) — broad conformational spreads around a fixed receptor. AlphaFold-style ensembles are five near-duplicate high-confidence models with a moving receptor (which we re-frame by superposition); this is a real distribution shift and the most likely reason the score fails to rank them. Only 5 poses per ensemble is also thin for a ranking task, and results are single-seed. The one robust finding is that **GraphPep can rescue AF3's occasional catastrophic rank-1 errors**, but it cannot be applied blindly — for Protenix it costs accuracy.

6.4 InterPepRank

Not run. Code is on Bitbucket (isaakh94/interpeprank), a 2020-era TensorFlow/Spektral/Keras stack, and its node features require an HHblits search of uniclust30_2016_03 (~50 GB) for the receptor PSSM. Deferred to the university cluster. The evaluation harness (`rerank_eval.py --reranker interpeprank`) is ready; only the score files are missing.

7. Limitations and pitfalls

Dataset

1. **Small n (13).** Adequate for descriptive comparison; **not** for significance testing between methods. All method rankings here are indicative.

2. **Uneven method coverage.** Chai-1 and Protenix span all 13 targets; AF3 10,

AlphaFold-Multimer 6, Boltz-2 1 (web submission is manual; Boltz-2 needs a GPU we don't have). Per-method means in §5.1 are over different subsets and are not yet strictly comparable — only the Chai-1 vs Protenix comparison is complete.

3. **Receptor redundancy.** DnaK is 7/13 targets; melittin is the peptide in 2. Effective diversity is lower than $n = 13$; family-level averaging is advisable.

4. **7SAY is a mis-specified target** — the "receptor" is an engineered GCN4/M-protein crystallisation scaffold, not a physiological LL-37 partner. Its near-zero DockQ for every method should probably be excluded from headline statistics.

5. **Caution targets (5/13)** are scored over a 50–78 % peptide fragment; their DockQ values carry less weight and are flagged with *.

6. **Single seed / server-default sampling.** No estimate of sampling variance within a method.

7. **Assembly ambiguity.** Biological assembly 1 was used throughout; the seed spreadsheet did not specify assembly vs deposited entry.

Data-quality traps encountered (and how they were caught)

8. **D-amino-acid peptides crash DockQ silently** (Bio.PDB drops the all-non-standard chain → `KeyError`). Must be filtered before scoring (criterion B).

9. **Literal x in RCSB SEQRES** (non-proteinogenic residues) is written verbatim into every predictor's input file. The discovery screen only checked *resolved* residues; `prep_native.py` checks SEQRES and caught 4JWI / 7NEF (criterion C).

10. **Name-based AMP identification is unreliable.** 2HD4 and 4N6P were added as "lactoferricin-related" by name association; their citation titles show a proteinase-K inhibitor peptide and a small-molecule drug-binding study respectively (criterion D). AMP relevance must be verified against the entry's own citation.

11. **>50 % resolved ≠ scorable.** 4JWD has 7/14 residues modelled but they do not contact the receptor, so DockQ returns no interface at all (criterion E, added after the first DockQ run produced empty output for every method on 4JWD).

Tooling issues (documented for reproducibility)

12. **GraphPep v1.1 release is incomplete** — `bin/getseq.awk` (called by `pre.py`) is missing; supplied as `methods/graphpep/getseq.awk`. `GraphPep.sh` is not whitespace-safe (the repo path contains a space) — worked around by staging each job into a space-free directory.

13. **ESM-2 checkpoint integrity.** The `esm2_t33_650M_UR50D.pt` already on disk was truncated (~4 MB short, no zip central directory); re-downloaded from the fair-esm host. `fair-esm 2.0.0` is also incompatible with `torch ≥ 2.6` (`weights_only=True` default rejects the checkpoint) — patched to `weights_only=False`.

8. Reproducibility

- **Target set:** `configs/targets.yaml` (every included and excluded entry, with reason). Full discovery output: `results/discovery_candidates.csv`.

- **Pipeline:** `pipeline/` — ``prep_native`` → `make_inputs`` → `collect_predictions`` → `run_dockq`` → `report``; `objective 2`` adds `rerank_prep`` → `run_dockq`` → `rerank_eval`` (+ `graphpep_prep``).

- **Scores:** `results/summary.csv` (per-pose DockQ), `results/confidences.csv` (tool confidences), `results/REPORT.md`, `results/rerank.md`, `results/rerank.csv`.

- **External:** DockQ v2 (`pip install DockQ`); GraphPep (Zenodo 10.5281/zenodo.17099863, weights bundled); ESM-2 `esm2_t33_650M_UR50D`; RCSB PDB accessions as listed.

- **Predictor versions/dates:** record per submission (COSMIC² pipeline version; AlphaFold Server date; Chai-1 / Protenix server version) — *to be tabulated*.